Hongyu (Victoria) Chen

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Education

University of California, Davis Jan 2022 – April 2027 (expected)
Ph.D. Candidate in Computer Science
Research: Machine Learning for Weather & Geospatial Data, High-Performance Computing, Computational Geometry

Georgia Institute of Technology Aug 2019 – Dec 2021
M.S. Computational Science & Engineering; M.S. Environmental Engineering GPA 3.89/4.0

Guangdong University of Technology Sep 2015 – Jun 2019
B.E. Environmental Engineering

Research Experience

Applied ML for Data-Limited Seasonal Prediction 2026 – Present
Cross-domain representation learning for Sierra Nevada snowpack · Manuscript in preparation

- **Originated and lead an applied ML research project** on data-limited snowpack prediction, translating observational scarcity, target reliability, and simulation-to-observation mismatch into **tractable research questions, baselines, and controlled experiments**; established an $R^2 = 0.39$ statistical baseline; **mentor two undergraduate researchers**.
- **Designed and evaluated CNN, attention-based, and Transformer models in PyTorch** to test competing hypotheses about spatial structure, long-range dependencies, and cross-domain representation transfer; validated simulation source domains against observed spatial and seasonal structure, with representation probes and failure analysis guiding each iteration.
- **Adapt and evaluate pretrained weather-AI models on NERSC Perlmutter**, including probabilistic NeuralGCM ensembles, long autoregressive ACE2-ERA5 rollouts, and **fine-tuning ACE2-ERA5 for downstream snowpack prediction**.

UXarray (DOE/NCAR Collaboration) 2022 – Present
github.com/UXARRAY/uxarray
- **Lead UXarray's spherical-geometry algorithm research and development**, with **primary ownership of the geometry stack** in an open-source Python framework for scalable geospatial analysis on unstructured meshes used in **production scientific workflows** (**200+ GitHub stars**, 50+ releases); **mentor NCAR summer interns since 2022**.
- **Designed robust, vectorized spherical-geometry algorithms** (intersections, spatial bounds, point-in-polygon) that form the basis of UXarray's production geometry stack, achieving **near machine-precision accuracy with no measurable intersection-kernel overhead** and enabling later optimizations of up to **88× lower peak memory** and **2.1× faster screening**.
- **Guided the design and integration of UXarray's LLM-agent (MCP) workflows** for remote HPC execution, exposing its geometry algorithms as agent-callable tools.

AccuSphGeom: Research Software for Accurate Spherical Geometry 2026 – Present
github.com/hongyuchen1030/AccuSphGeom

- **Translated two first-author spherical-geometry/regridding algorithms into an open-source C++17 library**, combining compensated arithmetic with SIMD/OpenMP to achieve **roughly double-working-precision accuracy at native FP64 throughput** under vectorized parallel execution.

TempestExtremes: Distributed Earth-System Data Processing 2022 – Present
github.com/ClimateGlobalChange/tempestextremes

- **Scaled a climate-analysis tool** from non-distributed C++ execution to **MPI distributed-memory processing**, enabling **300+ GB Earth-system workloads across 2,000+ CPU cores** on NERSC Perlmutter.

Work Experience

ADP Summer 2021
Software Engineering Intern Alpharetta, GA
- Developed a full-stack **inventory management system** (PostgreSQL, React, **RESTful Node.js APIs**); resolved **release-blocking issues**, reducing downtime by **30%**.

Geonamic Systems Inc. Spring 2021
Software Developer Intern Duluth, GA
- Developed production **geospatial data systems** in **C#, Python, and SQL**; optimized SQL queries and database operations, reducing application latency by **20%**.

Technical Skills

Machine Learning: Python, PyTorch, JAX, TensorFlow, scikit-learn; Transformers, CNNs, Attention; Representation Learning, Transfer Learning, Fine-Tuning, Generative/Autoregressive Models, Model Evaluation
Weather & Geospatial Data: Xarray, Dask, Zarr, CDO/NCO; gridded & unstructured-mesh data, spherical geometry, regridding
High-Performance Computing: CUDA, MPI, OpenMP, SIMD/Vectorization; distributed computing on NERSC Perlmutter
Software Engineering: C++, Python, Java, SQL; Git, CMake, Docker, REST APIs

Selected Publications & Honors

- **H. Chen** (first author) et al., *Fast and Accurate Intersections on a Sphere*, **SIAM Journal on Scientific Computing**, 2026.
- **H. Chen** (first author) et al., *Accurate and Robust Geometric Algorithms for Regridding on the Sphere*, **Geoscientific Model Development**, 2026.
- *Learning Across Timescales and Domains: Representation Transfer for Sierra Nevada Snow Water Equivalent Prediction*, **manuscript in preparation**, 2026.

Presentations & Workshops: WCRP Digital Earths Global km-Scale Hackathon, U.S.-West Berkeley Node, 2025 (**Workshop Lead: Processing HEALPix Data with UXarray**);
Joint Mathematics Meetings, 2025 (**Invited Speaker**); SIAM Conference on Mathematical & Computational Issues in the Geosciences, 2025; Scientific Computing with Python Conference (SciPy), 2023; American Geophysical Union Annual Meeting, 2023.
Honors: AGU Outstanding Student Presentation Award, 2023; UC Davis CS Summer Research Fellowship, 2023.